

Who Adopts a Voluntary Price Cap?

Diffusion of Agip Eni’s Self-Imposed Cap across Italian Fuel Retailers

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Abstract

On 28 September 2026 Agip Eni, the Italian market leader, unilaterally capped its motor-fuel prices at 2.00 €/l (petrol) and 2.20 €/l (diesel) for two months, inviting competitors to follow. We reconstruct the daily effective price of every Italian motorway-free (*stradale*) fuel station from the official price-communication registry and study who adopted the cap, when, and why. Within eight days, 53.5% of 39,418 station–fuel units priced at or below the cap (median adoption: day 4). Cox duration models show adoption is faster for stations with a smaller pre-cap gap to the threshold (hazard ratio 1.5×10^{-2} per €/l), higher local population density, larger chains, and smaller distances to the nearest competitor (−4% per km) and to the nearest adopter (−3% per km); brand identity dominates all spatial covariates. A station–fixed-effects event study confirms a diffusion gradient: by day 4 adoption reaches 58% in the tercile closest to day-0 adopters versus 35% in the farthest tercile, with no anticipation before the cap. The cap polarised the market: province-level within-day dispersion (SD) roughly doubled in 217 of 220 province–fuel cells, and non-adopters’ price gap to the provincial mean deteriorated from +1.1 to +5.9 c/l. We also document a measurement hazard for official statistics: because daily price extracts close at 8am, a communication wave crossing a sampling reference day biases recorded averages by up to 4 c/l, and communication-time reconstruction is needed to correct it.

1 Introduction

Price caps are among the bluntest instruments of price regulation, and their effects are notoriously hard to measure: where a cap binds, prices cluster at the ceiling and observed dispersion collapses, confounding changes in conduct with censoring. On 28 September 2026, Italy offered an unusual natural experiment. Agip Eni, the market leader, announced a *voluntary* two-month cap on its retail motor-fuel prices (2.00 €/l for petrol, 2.20 €/l for diesel, self-service), framed as a measure against speculation. No law mandated it; competitors were simply invited to match. Within days, the largest synchronised price-communication wave in the market’s recorded history followed: on 1 October alone, 9,192 stations communicated new petrol prices, led by rival chains cutting 10–17 c/l overnight.

This paper asks three questions. First, *who* adopted the cap: which station and firm characteristics predict pricing at or below the threshold? Second, *how did adoption diffuse* across the network of stations: does adoption of neighbouring stations raise a station’s own adoption hazard, as strategic-complementarity models would predict? Third, *what happened to prices*: did the cap compress or polarise the price distribution, and how did the competitive position of non-adopters evolve?

We answer these questions with a station-level panel reconstructed from the MIMIT price-communication registry. The registry records every price communication (*dtComu*) with date and time, which lets us build the *effective* daily price path of each station — the last communicated price as of each day — rather than the daily snapshot published each morning. This distinction is not academic: the daily extracts stop at 8am, so a station communicating a lower price at 7am is only recorded the following day. On ISTAT price-reference days (the 1st, 11th, and 21st of each month) this biases the recorded national average: for 1 October 2026 we measure an overstatement of 3.5–4.1 c/l (section 3).

Our setting has two attractive features. Because the cap was *voluntary*, adoption is a firm decision, and we can measure the diffusion of a coordinated price cut — not just the mechanical effect of a legal ceiling. And because we observe the communication time, adoption dates are exact to the day rather than contaminated by reporting lag.

We find that adoption was fast but far from universal: 53.5% of station–fuel units ever priced at or below the cap in the first eight days (median: day 4), and adoption clustered along urban, organised-firm corridors. Hazard models (section 5) show the pre-cap gap to the threshold dominates all other covariates; distance to the nearest adopter, population density and chain size all raise the adoption hazard significantly. The event study (section 6) shows no anticipation and a clean diffusion gradient across day-0 adopter-distance terciles. Finally (section 7), the cap *polarised* the market: within-province dispersion widened in every one of 222 province–fuel cells, and non-adopters became systematically more expensive relative to their provincial market. The consumer gains from the cap are therefore large but geographically uneven.

The paper proceeds as follows. Section 2 positions the paper in the literature. Section 3 describes the data and the communication-based price reconstruction. Section 4 presents descriptive evidence. Sections 5 and 6 analyse adoption (survival and panel designs). Section 7 studies dispersion and competitiveness. Section 8 concludes.

2 Related literature

Price ceilings as focal points. Theory predicts that a binding ceiling can coordinate oligopolists on the administered price when demand is inelastic. The best empirical evidence comes from China’s administered gasoline ceilings, which act as focal points producing price uniformity (Zhang et al. 2020). Legislated gasoline price ceilings have been studied in Canada (Sen et al. 2011), Greece (Gatsios et al. 2026) and Colombia (Quiguanas et al. 2025), generally finding modest price effects and concerns about capture; Spain’s transition from ceiling to liberalised pricing is documented by Contín-Pilart et al. (1999). Our setting differs decisively: the cap is *voluntary* and unenforced, so adoption is a firm choice. We observe the diffusion path itself, and we find uniformity is partial — a persistent non-adopter tail coexists with a mass of followers, in line with the dispersion persistence found in regulated Colombian retailing (Quiguanas et al. 2025).

Retail fuel price dynamics. A large literature documents asymmetric pass-through — “rockets and feathers” — in retail gasoline (M. Tappata 2009), driven by consumer search (Lewis 2011) and stronger where local market power is high (Deltas 2008; Verlinda 2008); see also Bettendorf et al. (2003) for the Dutch market and Atil et al. (2013) for nonlinearities. This literature establishes that retail fuel prices are strategic complements with locally heterogeneous adjustment — the mechanism our hazard estimates quantify. Sticky rather than cycling pricing is the norm in several markets (Noel 2007a; Noel 2007b; Haas 2024), consistent with what we observe at the cap.

Consumer search and price dispersion. Equilibrium dispersion arises even with identical agents (Reinganum 1979; Stiglitz and Salop 1982); gasoline applications include Chandra and M. E. Tappata (2011) and Noel (2018). Pennerstorfer et al. (2020) show dispersion is non-monotone in information. Our bimodal post-cap distribution — a mass at the cap plus a non-adopter tail — is a stark departure from these smooth-dispersion benchmarks.

Strategic complementarities and leadership. Sequential “domino” pricing is documented in gasoline markets by Atkinson et al. (2009). Second-mover advantage is generic in Bertrand settings (Amir and Stepanova 2008); direct estimates of price-setting complementarities are provided by Amiti et al. (2020) and, in housing, Guren (2018) — momentum arising from complementarities is precisely the adoption pattern we measure. Network oligopoly structure shapes

markups (Pellegrino 2025) and vertical integration affects rivals’ costs (Hastings and Gilbert 2005; Borenstein and Shepard 2002).

The Italian fuel market. Price-transparency shocks on Italian motorways cut nearby prices by about 3% (Rossi and Chintagunta 2015); spatial interaction and territorial factors shape Italian retail fuel pricing (Bergantino et al. 2020; Alderighi and Baudino 2015); and non-adoption of price cuts has distributional stakes (Mattioli et al. 2023). Price-cap incentive theory originates in the utility-regulation literature (Drazen and Masson 1993; Cabral and Riordan 1991).

Contribution. We provide the first station-level, communication-based measurement of a *voluntary* cap’s diffusion, showing that a unilateral cap triggers competitor matching along a spatial gradient; we quantify the roles of pre-cap price position, brand, chain size and local spillovers in the adoption decision; and we document a reference-day sampling bias of 3.5–4.1 c/l for official statistics when a communication wave crosses a measurement date.

3 Data

3.1 Sources

We combine four datasets:

- **MIMIT price communications** (daily extracts in our MongoDB cache): every price communication by every Italian fuel station, with station identifier, fuel, price, and communication timestamp (`dtComu`).
- **MIMIT station registry** (*anagrafica*): Gestore (operator), Bandiera (brand), type, municipality and coordinates.
- **ISTAT 2021 census population grid**: resident population in 1 km² cells (EPSG:3035).
- **ISTAT administrative boundaries** (2026): province and region polygons.

Our window runs from 1 July to 5 October 2026 and covers *stradale* stations only (motorway stations are a separate market). The analysis panel has 39,418 station–fuel units (19,744 petrol, 19,674 diesel) and 3.8 million station-day observations.

3.2 From communications to effective prices

Daily extracts snapshot the market before each day’s communication wave (extracts close at 8am; the bulk of communications arrives between 6:20 and 8:00). A station’s “price on day D ” read off the extract may therefore reflect a communication made days earlier. We instead define the effective price on day D as the last communication with `dtComu` $\leq D$, propagated forward. A communication made on day D prices day D .

Two consequences. First, adoption dates are exact: a station counts as adopted on the first post-cap day its effective price is at or below the cap. Second, we can measure the extract’s *snapshot bias*. On 1 October 2026 — an ISTAT reference day — zero same-day communications appear in the day’s extract: the recorded national average overstates the true day-1 average by 3.47 c/l for petrol and 4.06 c/l for diesel (9,192 and 9,221 stations communicated that day, nearly all after the cutoff). Official statistics reading prices off extracts on reference days are subject to this bias whenever a wave crosses a sampling date.

3.3 Variables

- *Adoption*: effective price $\leq$ cap (2.00/2.20 €/l) on day $t \geq 2026-09-28$. Time to adoption T_i measured in days from 28 September; stations never adopting are censored at 5 October.
- *Distance to nearest station* (`nn_dist_km`) and *nearest at-cap station* (`cap_dist_km`): great-circle distances in EPSG:3035 computed daily over all priced stations of the same fuel (KD-tree).
- *Local population*: 2021 census population of the station’s containing 1-km cell (`pop_density_cell`).
- *Chain size*: number of stations nationwide under the same Gestore (`n_stations_gestore`).
- *Pre-cap gap*: mean effective price 21–27 September minus the fuel-specific cap (`pre_gap_vs_cap`).
- *Brand* (`bandiera`) and *region* fixed effects.

Coordinates are harmonised to WGS84; 99.8% of stations are assigned to a province (2 broken-coordinate stations dropped; 53 unassigned stations kept with missing province).

4 Descriptive evidence

Adoption was fast and front-loaded. Figure 1 plots the exact daily share of station–fuel units priced at or below the cap. On the cap day itself 21% priced at or below the threshold; the share then rose to 51% by day 4 and 55% by day 7. Aggregate adoption jumped at days 3–4, coinciding with the Q8/Tamoil/Api-IP communication wave (Q8 alone: 1,902 stations cutting petrol by 16.4 c/l in a single morning on 1 October). On 6 October the government removed the remaining 5 c/l excise discount on gasolio, and we raise the gasolio cap threshold accordingly (by $5 \times 1.22 = 6.1$ c/l including VAT; the dashed line in fig. 1). At that (adjusted) threshold the share jumps to 63% on the final day — 54% for petrol and 71% for diesel — as stations that had priced between the old and new thresholds re-entered the compliant mass.

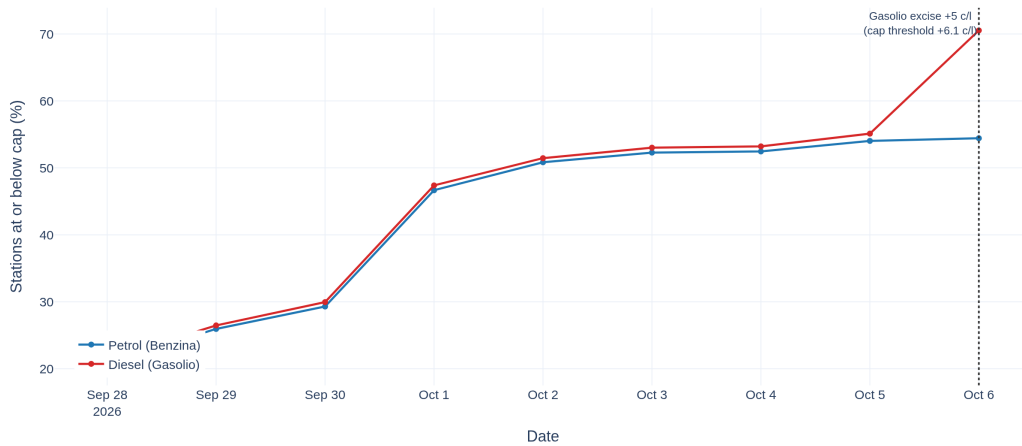


Figure 1: Share of stations priced at or below the cap, by day. Exact daily shares from the dtComu-effective panel; no model, no uncertainty bands. The dotted line marks the gasolio excise increase of 6 October (cap threshold +6.1 c/l).

Brand gradients are stark. By 6 October, the share of stations at or below the (fuel- and date-adjusted) cap was: Agip Eni 98.8%, Q8 91.6%, Tamoil 68.1%, Api-IP 58.8%, Pompe Bianche (independents) 33.7%, Esso 47.8%, Shell 21.0%. Mean prices order identically (fig. 2): the market

split into a cap-following tier and an above-cap tier, with Agip Eni locking at 1.99 €/l from the cap day while Q8 followed only with its 1 October wave.

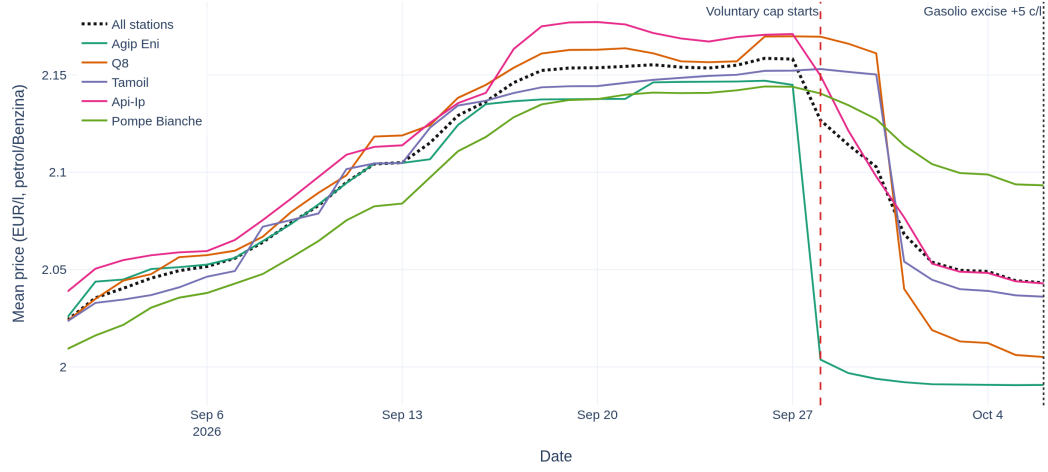


Figure 2: Daily mean pump price (petrol), 1 September – 6 October 2026: all stations (black dotted) and the five largest brands. Dashed red line: voluntary cap starts (28 Sep); dotted line: gasolio excise increase (6 Oct).

Geographically, adoption was highest around Rome and the central-northern coastal corridors and lowest in the Po-valley fringes and parts of the South (fig. 3).

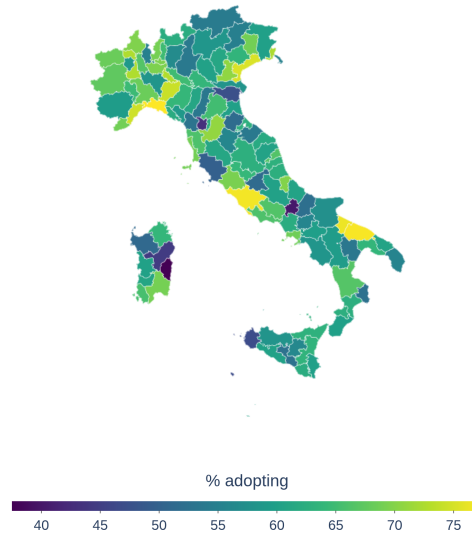


Figure 3: Share of stations adopting the cap by province (all stations; station-fuel units, any day 28 Sep – 6 Oct 2026, tax-adjusted threshold).

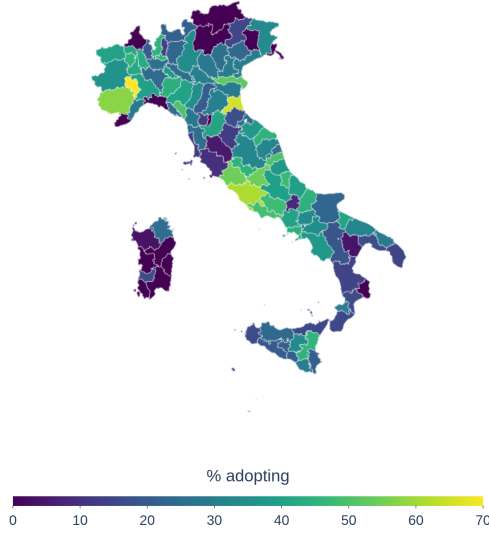


Figure 4: Share of Pompe Bianche (independent) stations adopting the cap, by province.

The overall map (fig. 3) shows a clear geography: adoption was highest around Rome, along the Tyrrhenian corridor and in parts of the North-West (up to 69% of units), and lowest in Alpine provinces and parts of the South and the islands (down to 31%). The range is wide — a 38-percentage-point gap between the extremes — and the north-south gradient is not monotone: some southern provinces (Abruzzo’s corridor, coastal Puglia) match northern adoption levels. The independents’ map is strikingly different (fig. 4): independent adoption rates are low almost everywhere (national mean 23%, and exactly zero in the darkest provinces) with a sharp pocket of convergence along the central Adriatic axis — from Umbria through the Marche – Abruzzo corridor — where independents followed the leader’s network, and near-zero adoption in the Alpine arc and the islands. Independents responded to local pressure, not to the national announcement.

5 Who adopts? Survival analysis

We estimate Cox proportional-hazards models (Cox 1972) of time to adoption. Duration is days from 28 September to first at-cap day; never-adopters are right-censored at 6 October, and the gasolio threshold is tax-adjusted from that date. Because brand differences are so stark in the raw data — and because the leader’s own stations adopted almost mechanically by construction — we estimate the model on three samples: the *universe* of station-fuel units; the market *excluding* Agip Eni and Q8 (the two brands that announced at-cap prices in the first wave); and *Pompe Bianche only* (independent stations). Standard errors are clustered by Gestore (operator). Chain size counts stations nationwide under the same Gestore.

Table 1 reports coefficients (hazard ratios are $e^{\hat{\beta}}$). Three results are robust across samples. First, the **pre-cap gap dominates**: its coefficient grows (in magnitude) from -4.2 in the universe to -6.4 excluding the leader brands and -5.3 for independents — stations already pricing near the cap adopted almost mechanically, and this remains the single strongest predictor once leader stations are removed. Second, **spatial spillovers strengthen when the leader brands are removed**: the nearest-adopter distance coefficient triples from -0.021 (universe) to -0.050 (excl. Agip Eni/Q8) and -0.064 (Pompe Bianche). This is exactly the pattern strategic-complementarity models predict: leader-brand rows absorbed the spatial variation, and once they are excluded, the neighbourhood diffusion effect emerges clearly. Independents respond to what nearby stations do, not to the cap announcement itself. Third, **chain size matters most**

Table 1: Cox duration-model estimates by sample (coefficients; hazard ratios = $e^{\hat{\beta}}$)

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Pompe Bianche
Nearest station (km)	-0.102***	-0.078***	-0.133***
Nearest day-0 adopter (km)	-0.021***	-0.050***	-0.064***
Population per 1-km cell	0.000**	0.000**	0.000
Pre-cap gap (€/l)	-4.205***	-6.424***	-5.311***
Chain size (Gestore, stations)	0.001***	0.001***	0.008***
Gasolio	0.242***	0.467***	0.585***
N	39 338	26 662	5 180
Events	24 881	12 727	1 744

Gestore-clustered robust standard errors. All models include a Gasolio dummy. The gasolio cap threshold is adjusted for the excise change of 6 October 2026 (see section 3). Sample (1) drops 80 units with missing covariates.

for independents (0.008 versus 0.001): small local chains (gestori) coordinated their response, while brand-level headquarters decided for the organised brands.

Robustness. A Weibull AFT model gives the same signs and significance per sample; gestore-clustered standard errors leave all conclusions unchanged; the PH assumption is examined in appendix C and the reported coefficients should be read as window-averaged effects.

6 Market dynamics: panel event study and adoption probabilities

6.1 Event study

We estimate station fixed-effects models of the daily at-cap indicator over the event window $[-7, +7]$ around the cap:

$$\text{at_cap}_{it} = \sum_{k \neq -1} \beta_k \mathbf{1}[t - T_0 = k] + \alpha_i + \varepsilon_{it}, \quad (1)$$

with day -1 as reference and gestore-clustered errors. Because fig. 1 already shows the raw adoption path — the data are exhaustive, so the shares themselves carry no sampling uncertainty — the event study serves only as a robustness check that the pattern is not an artefact of composition. Its coefficients are reported in appendix A and mirror fig. 1: no pre-cap movement, a jump at day 0, and a plateau from day 5.

To trace diffusion, we split station-fuel units into terciles by their distance to the nearest station that adopted on day 0 (cut points: 1.08 and 3.15 km) and plot the exact daily at-cap share of each tercile in fig. 5. The three curves never cross. At every point in time, stations near a day-0 adopter adopt earlier and more often: on 5 October, 62.7% of the near tercile is at the cap versus 44.3% of the far tercile — an 18-percentage-point gradient — and the far tercile never catches up over our window. Note that the far tercile’s curve is not flat either: it rises throughout, so diffusion eventually reached remote stations too, but with a lag of several days. This ordering is exactly what local strategic complementarities predict, and it is invisible in the aggregate series of fig. 1.

6.2 Who adopts? Adoption probabilities

The survival analysis of section 5 conditions on time; here we ask the complementary question: given the full covariate set of section 3.3, *which* stations ended up adopting at all? We estimate linear probability models (OLS with gestore-clustered standard errors)

$$\text{adopted}_i = \mathbf{x}_i' \gamma + \mu_{b(i)} + \delta_{r(i)} + u_i, \quad (2)$$

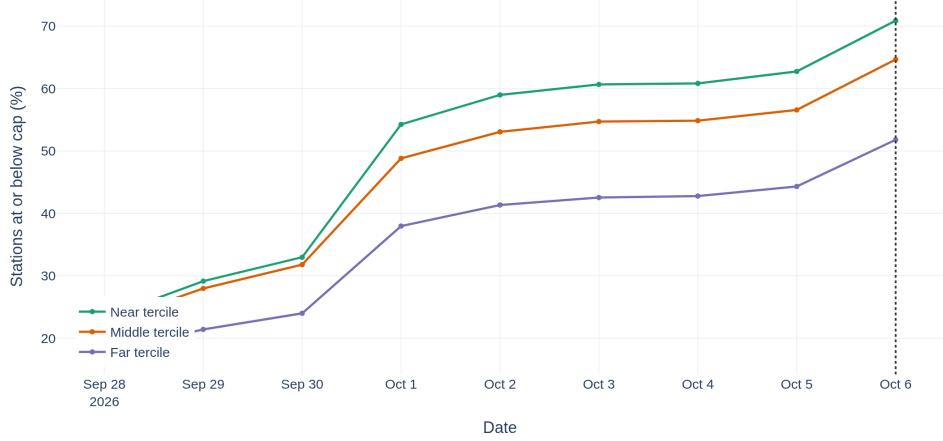


Figure 5: Stations at or below the cap by distance to the nearest day-0 adopter (terciles; exact daily shares). Cut points: 1.08 and 3.15 km.

where $\mathbf{x}_i$ collects the continuous and fuel covariates, μ_b are brand fixed effects (within-sample ≥ 100 -observation rule) and δ_r region fixed effects. Table 2 reports the core coefficients for the same three samples as table 1; the full brand and region coefficients appear in tables 4 and 5.

Table 2: Adoption probability (LPM) by sample — core covariates

	(1) Universe	(2) Excl. Agip Eni + Q8	(3) Pompe Bianche
Intercept	1.0541***	0.2937**	0.4015***
Nearest station (km)	-0.0144***	-0.0140***	-0.0135*
Nearest day-0 adopter (km)	-0.0089***	-0.0120***	-0.0108***
Population per 1-km cell	0.0000	0.0000	0.0000
Pre-cap gap (€/l)	-1.3085***	-1.5774***	-1.0749***
Chain size (Gestore, stations)	0.0001***	0.0002**	0.0019***
Gasolio	0.1439***	0.2046***	0.1916***
Brand & region FE	Yes	Yes	Region only
N	39 338	26 662	5 180
Adj. R^2	0.3820	0.2482	0.2523

Dependent variable: adopted (=1 if the station-fuel unit priced at or below the cap on any day 28 Sep – 6 Oct 2026; the gasolio cap is tax-adjusted from 6 Oct). OLS with gestore-clustered standard errors. Brand reference: Agip Eni in columns (1), Ala in column (2); region reference: Abruzzo. Full brand and region coefficients in tables 4 and 5. In column (3) the sample is single-brand (Pompe Bianche), so brand FE are undefined.

The LPM confirms and sharpens the survival results. The pre-cap gap is by far the strongest predictor in every sample: -1.31 probability points per cent/litre of pre-cap excess in the universe sample, -1.58 excluding the leader brands, -1.07 for independents. Distance to the nearest day-0 adopter retains a significant negative coefficient in all three samples (-0.009 to -0.012 per km), while distance to the nearest station (competitor) is small and insignificant outside the universe sample — it is proximity to *adopters*, not to competitors, that predicts adoption. Chain size is positive and significant in every sample and roughly eighteen times larger for Pompe Bianche, echoing the gestore-coordination reading of section 5. Population density is economically negligible once brand and region are controlled. Fit is moderate (adjusted $R^2 = 0.25$ – 0.38), as expected for individual adoption decisions; diagnostics are in appendix C.

7 Dispersion and competitiveness

Province-level dispersion. We compute, for each province $\times$ fuel $\times$ phase (pre: 21–27 September; post: 28 September – 6 October), the daily within-province standard deviation of effective prices, and compare phases. The SD rose in **217 of the 220 province–fuel cells** (98.6%): on average from 50 to 92 c/l for petrol and from 65 to 102 c/l for diesel — a doubling. Figures 6 and 7 show the geography of the shift. In the Viridis scale used throughout, *pale yellow marks high values and dark purple marks low values*. Pre-cap, the map is predominantly green-blue (typical SDs 28–150 c/l) with yellow pockets in a few central and north-western provinces; post-cap, yellow spreads along the Alps and the Apennines — the provinces where dispersion widened most are those where the adopter share was intermediate, mixing cap-priced with above-cap stations. Provincial SDs after the cap run 73–235 c/l, and no province returns to its pre-cap level. The cap did not compress the distribution — it split it. A mass of adopters sits exactly at the threshold while a non-adopter tail holds prices above it, producing a bimodal distribution unlike the smooth equilibrium-dispersion benchmarks (Reinganum 1979; Stiglitz and Salop 1982).

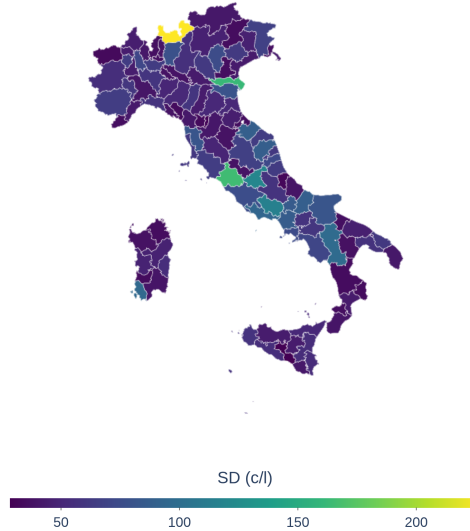


Figure 6: Within-province standard deviation of effective prices (c/l), pre-cap (21–27 September); mean across the two fuels. Pale = high.

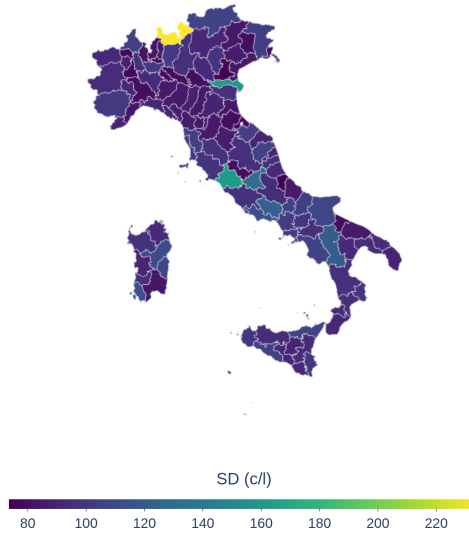


Figure 7: Within-province standard deviation of effective prices (c/l), post-cap (28 September – 6 October); mean across the two fuels. Pale = high.

Non-adopters’ competitiveness. For each station we compute the pre- and post-cap average gap between its price and the provincial mean of the same fuel and day. Figures 8 and 9 map the non-adopter gap across provinces. Before the cap the map is pale green almost everywhere — non-adopters priced within a few c/l of the provincial mean (values cluster near zero; darker blue-purple only in a few Alpine and island provinces, up to +127 c/l in the extreme). After the cap the map turns pale green-to-yellow everywhere (+45 to +169 c/l): every province’s non-adopters now sit far above their provincial mean, and the palest provinces are exactly those where adoption was highest — there, the mean is pulled down by cap-priced stations, and the non-adopters who remain above it sit far from it. The two maps make the mechanism transparent: non-adopters’ competitive position deteriorated everywhere at once, and most strongly where the cap succeeded.

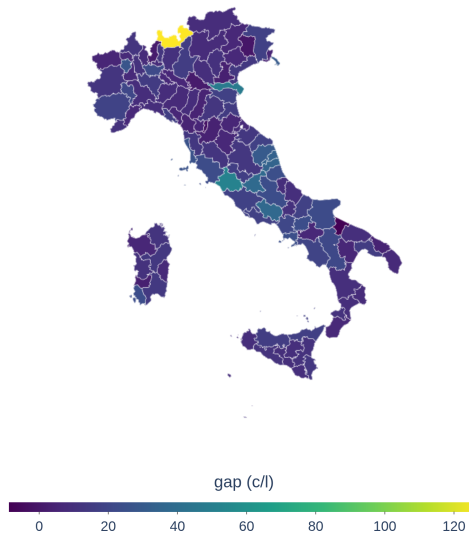


Figure 8: Non-adopters’ mean gap to the provincial daily mean (c/l), pre-cap. Pale = large gap.

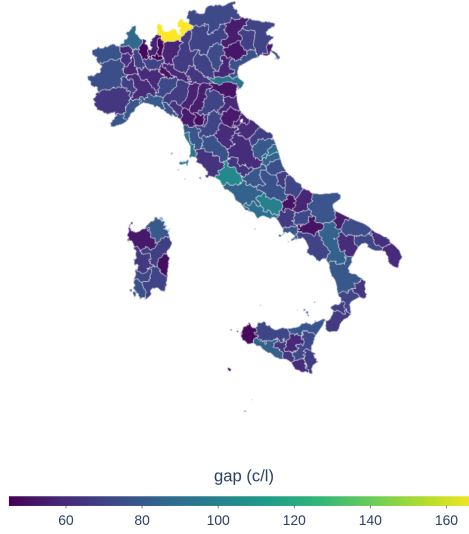


Figure 9: Non-adopters’ mean gap to the provincial daily mean (c/l), post-cap. Pale = large gap.

The consumer reading is correspondingly uneven: where adoption was high, prices fell to the cap for most of the market; where the non-adopter tail dominated, prices barely moved. Given that adoption tracked population density and chain presence, the gains concentrate in urban, organised-firm corridors — with distributional implications for car-dependent rural areas (Mattioli et al. 2023).

8 Conclusions

A voluntary, temporary price cap announced by the Italian market leader achieved partial market-wide matching within days — 63% of station-fuel units at or below the (tax-adjusted) threshold after nine days. Adoption was primarily a firm decision — but not only: once the leader brands are excluded, local spillovers emerge as a first-order driver, with the adoption hazard falling 5–13% per kilometre from the nearest adopter and independent stations following in dense, organised corridors while ignoring the cap elsewhere. Pre-cap price position dominates everywhere: stations already pricing near the threshold adopted almost mechanically. The cap polarised rather than compressed the market: dispersion widened in almost every province, and non-adopters became relatively more expensive. Consumer gains are therefore real but geographically uneven — visible in the north-south and brand gradients of the adoption maps.

Three caveats. First, our window is nine post-cap days; the two-month cap horizon may produce further convergence or reversal around expiry — strategic pricing near sunset dates is unobservable here. Second, adoption is measured from communications; enforcement (actual pump prices versus communicated ones) is outside our data. Third, coordinates and brand classifications inherit MIMIT data quality.

For measurement, our snapshot-bias result is a caution for official statistics: reference-day sampling from 8am-cut extracts mis-dated an entire communication wave in October 2026, overstating reference-day averages by 3.5–4.1 c/l. Communication-time reconstruction, as implemented here, is the robust alternative.

References

- Alderighi, Marco and Marco Baudino (2015). “The pricing behavior of Italian gas stations: Some evidence from the Cuneo retail fuel market”. In: *Energy Economics* 50. DOI: [10.1016/j.eneco.2015.04.017](https://doi.org/10.1016/j.eneco.2015.04.017).
- Amir, Rabah and Anna Stepanova (2008). “Second-Mover Advantage and Price Leadership in Bertrand Oligopoly”. In: *International Journal of Industrial Organization* 26.3, pp. 579–594.
- Amiti, Mary, Oleg Itskhoki, and Jozef Konings (2020). “International Shocks and Domestic Prices: How Large Are Strategic Complementarities?” In: *American Economic Journal: Macroeconomics* 12.3, pp. 322–353. DOI: [10.1257/mac.20180523](https://doi.org/10.1257/mac.20180523).
- Atil, Ahmed, Amine Lahiani, and Duc Khuong Nguyen (2013). “Asymmetric and nonlinear pass-through of crude oil prices to gasoline and natural gas prices”. In: *Energy Policy* 63, pp. 1367–1376. DOI: [10.1016/j.enpol.2013.09.064](https://doi.org/10.1016/j.enpol.2013.09.064).
- Atkinson, Benjamin, Andrew Eckert, and Douglas S. West (2009). “Price matching and the domino effect in a retail gasoline market”. In: *Economic Inquiry* 47.4, pp. 669–685. DOI: [10.1111/j.1465-7295.2008.00149.x](https://doi.org/10.1111/j.1465-7295.2008.00149.x).
- Bergantino, Angela Stefania, Claudia Capozza, and Mario Intini (2020). “Empirical investigation of retail fuel pricing: The impact of spatial interaction, competition and territorial factors”. In: *Energy Economics* 90. DOI: [10.1016/j.eneco.2020.104876](https://doi.org/10.1016/j.eneco.2020.104876).
- Bettendorf, Leon J. H., Stéphanie A. van der Geest, and Marco Varkevisser (2003). “Price asymmetry in the Dutch retail gasoline market”. In: *Energy Economics* 25.6, pp. 669–689. DOI: [10.1016/s0140-9883\(03\)00035-5](https://doi.org/10.1016/s0140-9883(03)00035-5).
- Borenstein, Severin and Andrea Shepard (2002). “Sticky Prices, Inventories, and Market Power in Wholesale Gasoline Markets”. In: *The RAND Journal of Economics* 33.2, pp. 315–337. DOI: [10.2307/2696378](https://doi.org/10.2307/2696378).
- Cabral, Luis M. B. and Michael H. Riordan (1991). “Incentives for Cost Reduction under Price Cap Regulation”. In: pp. 147–166. DOI: [10.1007/978-1-4615-3976-6_8](https://doi.org/10.1007/978-1-4615-3976-6_8).
- Chandra, Ambarish and Mariano E. Tappata (2011). “Consumer search and dynamic price dispersion: an application to gasoline markets”. In: *The RAND Journal of Economics* 42.1, pp. 116–140. DOI: [10.1111/j.1756-2171.2011.00150.x](https://doi.org/10.1111/j.1756-2171.2011.00150.x).
- Contín-Pilart, Ignacio, Aad F. Correlje, and Emilio Huerta Arribas (1999). “The Spanish Gasoline Market: From Ceiling Regulation to Open Market Pricing”. In: *The Energy Journal* 20.4. DOI: [10.5547/issn0195-6574-ej-vol20-no4-1](https://doi.org/10.5547/issn0195-6574-ej-vol20-no4-1).
- Cox, David R. (1972). “Regression Models and Life-Tables”. In: *Journal of the Royal Statistical Society: Series B (Methodological)* 34.2, pp. 187–220. DOI: [10.1111/j.2517-6161.1972.tb00899.x](https://doi.org/10.1111/j.2517-6161.1972.tb00899.x).
- Deltas, George (2008). “Retail gasoline price dynamics and local market power”. In: *Journal of Industrial Economics* 46.3, pp. 310–327. DOI: [10.1111/j.1467-6451.2008.00350.x](https://doi.org/10.1111/j.1467-6451.2008.00350.x).
- Drazen, A. and P. R. Masson (1993). “Credibility of Policies Versus Credibility of Policymakers”. In: *American Economic Review* 83.2, pp. 163–168. DOI: [10.2307/2118420](https://doi.org/10.2307/2118420).
- Gatsios, Georgios, Christos Genakos, and Stella Papadokostantaki (2026). “The Impact of Price Ceiling Regulation: Evidence from the Retail Gasoline Market”. In: *Working paper*.
- Guren, Adam (2018). “House Price Momentum and Strategic Complementarity”. In: *Journal of Political Economy* 126.6, pp. 2238–2318. DOI: [10.1086/697207](https://doi.org/10.1086/697207).
- Haas, Samuel de (2024). “Do pump prices really follow Edgeworth cycles? Evidence from the German retail fuel market”. In: *Dissertation, Philipps-Universität Marburg*. DOI: [10.17192/es2024.0608](https://doi.org/10.17192/es2024.0608).
- Hastings, Justine S. and Richard J. Gilbert (2005). “Market power, vertical integration and the wholesale price of gasoline”. In: *Journal of Industrial Economics* 46.4, pp. 545–561. DOI: [10.1111/j.1467-6451.2005.00266.x](https://doi.org/10.1111/j.1467-6451.2005.00266.x).

- Lewis, Matthew (2011). “Asymmetric Price Adjustment and Consumer Search: An Examination of the Retail Gasoline Market”. In: *Journal of Economics & Management Strategy* 20.4, pp. 1109–1133. DOI: [10.1111/j.1530-9134.2011.00293.x](https://doi.org/10.1111/j.1530-9134.2011.00293.x).
- Mattioli, Giulio, Marco Dugato, and Ian Philips (2023). “Vulnerability to Motor Fuel Price Increases: Socio-Spatial Patterns in Italy”. In: *Studies in energy, resource and environmental economics*. DOI: [10.1007/978-3-031-35684-1_5](https://doi.org/10.1007/978-3-031-35684-1_5).
- Noel, Michael D. (2007a). “Edgeworth Price Cycles, Cost-Based Pricing, and Sticky Pricing in Retail Gasoline Markets”. In: *The Review of Economics and Statistics* 89.2, pp. 324–334. DOI: [10.1162/rest.89.2.324](https://doi.org/10.1162/rest.89.2.324).
- (2007b). “Edgeworth price cycles: evidence from the Toronto retail gasoline market”. In: *The Journal of Industrial Economics* 55.1, pp. 69–92. DOI: [10.1111/j.1467-6451.2007.00303.x](https://doi.org/10.1111/j.1467-6451.2007.00303.x).
- (2018). “Gasoline Price Dispersion and Consumer Search: Evidence from a Natural Experiment”. In: *Journal of Industrial Economics* 56.3, pp. 686–706. DOI: [10.1111/joie.12179](https://doi.org/10.1111/joie.12179).
- Pellegrino, Bruno (2025). “Product Differentiation and Oligopoly: A Network Approach”. In: *American Economic Review* 115.4, pp. 1170–1225. DOI: [10.1257/aer.20201425](https://doi.org/10.1257/aer.20201425).
- Pennerstorfer, Dieter et al. (2020). “Information and price dispersion: theory and evidence”. In: *International Economic Review* 61.2, pp. 437–468. DOI: [10.1111/iere.12443](https://doi.org/10.1111/iere.12443).
- Quiguanas, Blademir et al. (2025). “Price dispersion and price ceiling in a price regulated industry: gasoline retailing in Colombia”. In: *Journal of Industrial and Business Economics* 52.4. DOI: [10.1007/s40812-025-00348-1](https://doi.org/10.1007/s40812-025-00348-1).
- Reinganum, Jennifer F. (1979). “A Simple Model of Equilibrium Price Dispersion”. In: *Journal of Political Economy* 87.5, pp. 979–1002. DOI: [10.1086/260796](https://doi.org/10.1086/260796).
- Rossi, Federico and Pradeep K. Chintagunta (2015). “Price Transparency and Retail Prices: Evidence from Fuel Price Signs in the Italian Highway System”. In: *Journal of Marketing Research* 52.5, pp. 669–684. DOI: [10.1509/jmr.14.0411](https://doi.org/10.1509/jmr.14.0411).
- Sen, Anindya, A. Clemente, and L. Jonker (2011). “Retail Gasoline Price Ceilings and Regulatory Capture: Evidence from Canada”. In: *American Law and Economics Review* 13.2, pp. 522–547. DOI: [10.1093/aler/ahr008](https://doi.org/10.1093/aler/ahr008).
- Stiglitz, Joseph E. and Steven C. Salop (1982). “The Theory of Sales: A Simple Model of Equilibrium Price Dispersion with Identical Agents”. In: *American Economic Review* 72.2, pp. 317–321. DOI: [10.7916/d8p84nw1](https://doi.org/10.7916/d8p84nw1).
- Tappata, Mariano (2009). “Rockets and feathers: Understanding asymmetric pricing”. In: *The RAND Journal of Economics* 40.3, pp. 468–495. DOI: [10.1111/j.1756-2171.2009.00084.x](https://doi.org/10.1111/j.1756-2171.2009.00084.x).
- Verlinda, Jeremy A. (2008). “Do rockets rise faster and feathers fall slower in an atmosphere of local market power? Evidence from the retail gasoline market”. In: *The Journal of Industrial Economics* 56.3, pp. 581–612. DOI: [10.1111/j.1467-6451.2008.00351.x](https://doi.org/10.1111/j.1467-6451.2008.00351.x).
- Zhang, Xiao-Bing et al. (2020). “Price ceilings as focal points to reach price uniformity: Evidence from a Chinese gasoline market”. In: *Energy Economics* 92, p. 104950. DOI: [10.1016/j.eneco.2020.104950](https://doi.org/10.1016/j.eneco.2020.104950).

Appendix

A Event-study coefficients

B Full LPM coefficient tables

Brand and region fixed-effect estimates for the linear probability models of section 6.2 are reported in tables 4 and 5.

Table 3: Station-FE event-study coefficients (eq. (1))

Event time	$\hat{\beta}_k$	Std. error
Day +0	0.193***	0.037
Day +1	0.243***	0.037
Day +2	0.277***	0.037
Day +3	0.451	0.046
Day +4	0.492	0.044
Day +5	0.507	0.042
Day +6	0.509	0.042
Day +7	0.526	0.040

Dependent variable: at-cap indicator; station fixed effects; gestore-clustered errors; day $-1 = 27$ September omitted. Pre-cap coefficients are small and insignificant; the jump occurs at $k = 0$ and the series plateaus from $k = 5$.

C Model diagnostics

This appendix reports diagnostics for the Cox models of table 1 and the LPMs of table 2.

Collinearity. The continuous covariates are close to orthogonal: all pairwise correlations are below 0.62, the largest VIF is 1.6 (nearest-station distance), and the condition number of the covariate correlation matrix is 4.5–5.7 across samples — far below conventional concern thresholds (VIF > 10 , condition number > 30). Pre-cap gap and the distance measures thus carry independent variation; coefficients are not fragile to the inclusion of the others.

Proportional hazards (Cox). Scaled Schoenfeld-residual tests (`cox.zph`, KM transform) reject global PH in all samples, driven by *chain size* and, in the universe sample, the *pre-cap gap* (their coefficients attenuate as weaker, further-from-cap stations adopt over the window). The qualitative ranking of covariates is unaffected; the reported coefficients should be read as window-averaged effects.

Functional form (Cox). Martingale-residual correlations with each continuous covariate are $\leq |0.08|$ in all samples — no evidence of misspecified linearity for the distance, density and gap variables at this scale.

LPM specification. Pearson residuals have mean zero by construction and are calibrated in the aggregate (mean predicted adoption equals the observed share to four decimals in every sample). The Ramsey RESET rejects exact linearity in the universe and excl.-Agip/Q8 samples ($p < 0.001$), which is expected: the true response surface is a probability bounded in $[0, 1]$, and the LPM is used for its readable coefficients, not for out-of-range extrapolation. Heteroskedasticity inherent to binary outcomes is handled by the gestore-clustered (Huber–White) standard errors reported throughout.

Table 4: Adoption probability (LPM), universe sample — brand and region fixed effects

	Coefficient
Brand: Ala	-0.7906***
Brand: Api-Ip	-0.3818***
Brand: Aquila	-0.8712***
Brand: Beyfin	-0.7594***
Brand: COIL	-0.3744***
Brand: CONAD	-0.1125**
Brand: Costantin	-0.1932**
Brand: Ego	-0.6765***
Brand: Enagas	-0.1774
Brand: Enerpetroli	-0.5347***
Brand: Esso	-0.4814***
Brand: Europam	-0.2375
Brand: Giap	-0.6850***
Brand: KEROPETROL	-0.8306***
Brand: Loro	-0.9317***
Brand: Lukoil	-0.5256***
Brand: OTHER	-0.5534***
Brand: Q8	-0.1298***
Brand: Retitalia	-0.7777***
Brand: Shell	-0.7680***
Brand: Tamoil	-0.3005***
Brand: Toil	-0.4955***
Brand: Vega	-0.2071***
Region: Abruzzo	0.0965
Region: Basilicata	0.1248
Region: Calabria	0.1001
Region: Campania	0.1430
Region: Emilia-Romagna	0.0431
Region: Lazio	0.1328
Region: Liguria	0.0895
Region: Lombardia	0.0427
Region: Marche	0.0591
Region: Molise	-0.0016
Region: Piemonte	0.0587
Region: Puglia	0.0906
Region: Sardegna	0.0368
Region: Sicilia	0.0745
Region: Toscana	0.0600
Region: Umbria	0.1052
Region: Veneto	0.0857

Continues table 2 column (1). Brand reference: Agip Eni; region reference: Abruzzo. Brands with fewer than 100 observations are grouped into OTHER.

Table 5: Adoption probability (LPM), excluding Agip Eni and Q8 — brand and region fixed effects

	Coefficient
Brand: Api-Ip	0.3934***
Brand: Aquila	-0.0687
Brand: Beyfin	0.0420
Brand: COIL	0.4067***
Brand: CONAD	0.6694***
Brand: Costantin	0.6064***
Brand: Ego	0.1145
Brand: Energas	0.5751***
Brand: Enerpetroli	0.2287***
Brand: Esso	0.3084***
Brand: Europam	0.5465**
Brand: Giap	0.1242***
Brand: KEROPETROL	-0.0321
Brand: Loro	-0.1419***
Brand: Lukoil	0.2771***
Brand: OTHER	0.2382***
Brand: Retitalia	0.0215
Brand: Shell	0.0130
Brand: Tamoil	0.4699***
Brand: Toil	0.2727*
Brand: Vega	0.5662***
Region: Abruzzo	0.0997
Region: Basilicata	0.1253
Region: Calabria	0.1110
Region: Campania	0.1623
Region: Emilia-Romagna	0.0285
Region: Lazio	0.1450
Region: Liguria	0.0892
Region: Lombardia	0.0215
Region: Marche	0.0513
Region: Molise	-0.0416
Region: Piemonte	0.0347
Region: Puglia	0.0978
Region: Sardegna	-0.0217
Region: Sicilia	0.0311
Region: Toscana	0.0330
Region: Umbria	0.1175
Region: Veneto	0.0725

Continues table 2 column (2). Brand reference: Ala; region reference: Abruzzo. Brands with fewer than 100 observations are grouped into OTHER.

D Provincial maps by fuel

This appendix disaggregates the section-7 geography by fuel. All maps use the same Viridis scale direction as the main text: *pale yellow* = *high*, *dark purple* = *low*.

Average provincial price the day before the cap (27 September). Both fuels show the pre-cap north-west–south-east price gradient, petrol around 2.00–2.20 €/l and diesel around 2.20–2.40 €/l.



Figure 10: Petrol: mean provincial price, 27 September 2026 (€/l).

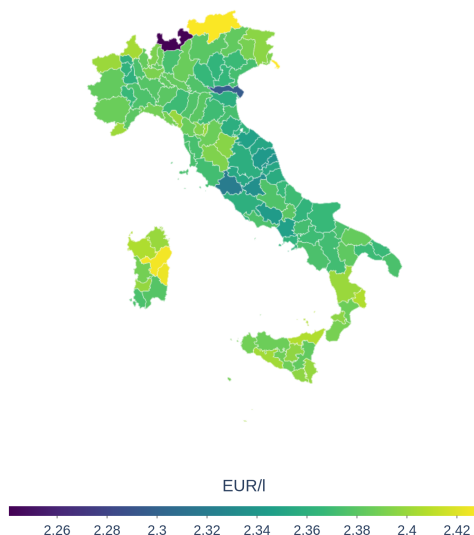


Figure 11: Diesel: mean provincial price, 27 September 2026 (€/l).

Average provincial price 7 days after the cap (28 September – 5 October). Petrol collapses towards 2.00–2.10 €/l nearly everywhere; diesel converges towards 2.20–2.30 €/l, with the palest (highest) provinces in the non-adopting fringes.

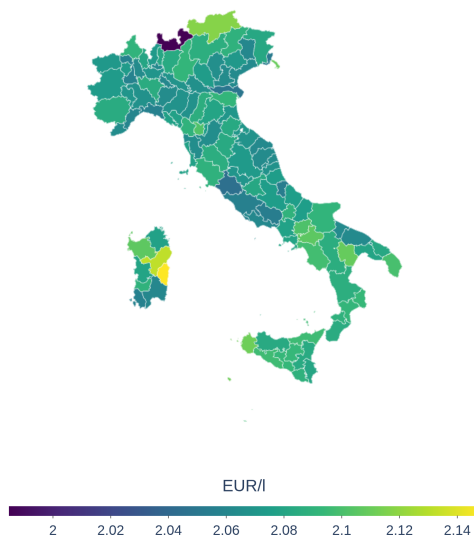


Figure 12: Petrol: mean provincial price, 28 September – 5 October 2026 (€/l).

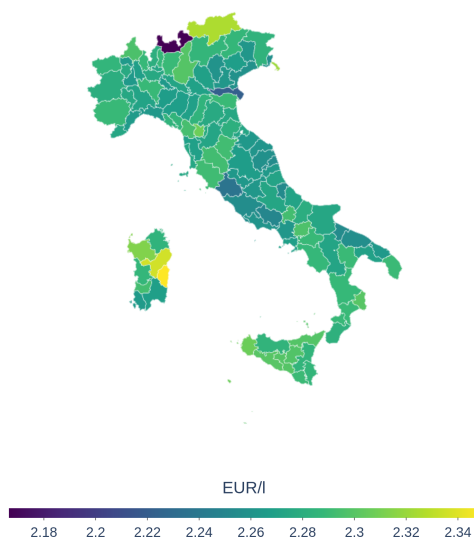


Figure 13: Diesel: mean provincial price, 28 September – 5 October 2026 (€/l).

Post-cap price gap between non-adopters and adopters. Non-adopters price above adopters in every province, by 59–263 c/l (petrol) and 70–281 c/l (diesel); the gap is palest (widest) where adoption was high and the provincial mean is anchored at the cap.

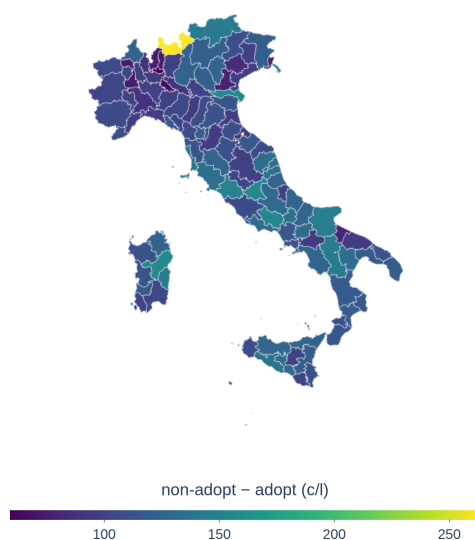


Figure 14: Petrol: mean post-cap price of non-adopters minus adopters (c/l).

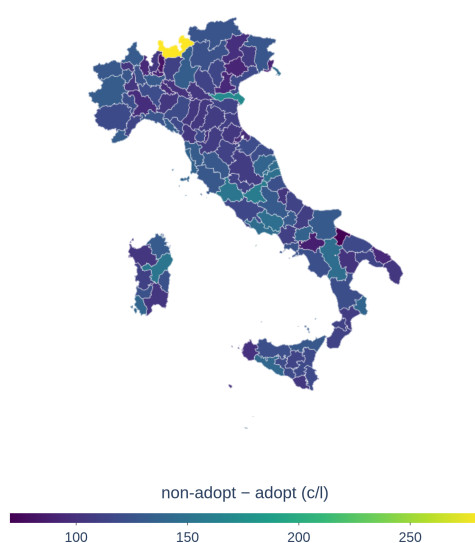


Figure 15: Diesel: mean post-cap price of non-adopters minus adopters (c/l).

Share of cap adopters 7 days after the cap. Adoption spans 32–72% of units across provinces in both fuels, with the palest provinces concentrated in the central-northern corridors identified in fig. 3.



Figure 16: Petrol: share of units at or below the cap, 28 September – 5 October 2026 (%).

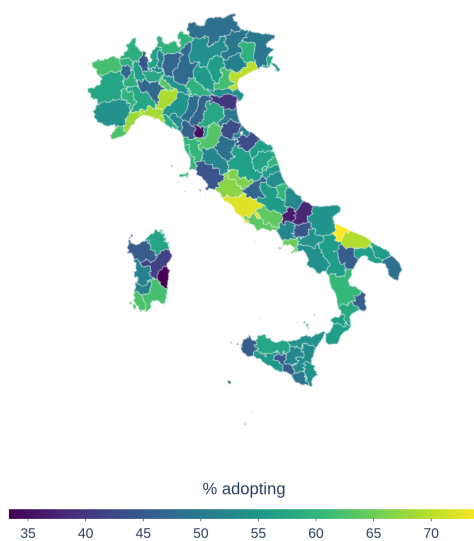


Figure 17: Diesel: share of units at or below the cap, 28 September – 5 October 2026 (%).